

Errata — *Wine Production Forecasting Using Rule-Based Machine Learning*

Hugo Filipe Queiróz Nogueira · MECD, FEUP · June 2026 — Corrections identified after submission of the version deposited for defense. Page numbers refer to the submitted PDF (*Tese Hugo Nogueira — final version*). All corrections are incorporated in the definitive version.

Erratum 1 · Page 36

LOCATION

Ch.3, §"CarenR: Distribution Rules via the CAREN Framework"

SUBMITTED TEXT

"CarenR is built on the CAREN association rule engine, **originally developed by Jorge et al. (2006)**"

CORRECTION

The CAREN engine was developed by **Azevedo (2003)**; Jorge, Azevedo & Pereira (2006) introduced the **distribution-rule** framework on top of it. Corrected to: "...originally developed by Azevedo (2003) and extended to the discovery of distribution rules in numeric data by Jorge et al. (2006)." New reference added: Azevedo, P. J. (2003), *CAREN — A Java based Apriori implementation for classification purposes*, Tech. Report, U. Minho.

Erratum 2 · Page 99

LOCATION

Bibliography, Jorge et al. (2006)

SUBMITTED TEXT

DOI 10.1007/11871637_25

CORRECTION

Correct DOI is 10.1007/11871637_26 (PKDD 2006, LNCS/LNAI vol. 4213, pp. 247–258, Springer). Volume and pages added.

Erratum 3 · Page 28 / 36–38

LOCATION

Ch.2 (p.28) and Ch.3 Step 3 (p.36)

SUBMITTED TEXT

Search described as **"beam search"** (Ch.2) and **"breadth-first ... most promising single conditions"** (Ch.3)

CORRECTION

CAREN performs an **Apriori-style levelwise search pruned by minimum support** (exhaustive over the support-frequent region, not a heuristic beam). Both passages harmonised to this description.

Erratum 4 · Page 28, 36, 38, 49, 100+ (Appx A/B)

LOCATION

Minimum support threshold

SUBMITTED TEXT

Stated as **"15%"** / " $\geq 15\%$ of observations" in Ch.2, Ch.3 (Step 5 and weighting paragraph), Appendix A and Appendix B

CORRECTION

Actual value used in the pipeline is **20%** (`carenr_min_sup <- 0.20` in `3_carenR_application.R` and `6_scenario_validation.R`), consistent with the results in Ch.4–6. All statements corrected to 20%. Derived counts corrected accordingly ("≈18 years out of 89"; weighting example "≈1.4×"). *This propagates a correction already logged in Appendix E to the chapters where it had been missed.*

Erratum 5 · Page 100

LOCATION

Bibliography, Moreira et al. (2022)

SUBMITTED TEXT

Title given as "**Contrast Association Rules for Extraordinary Nodes**", plus a printed "TODO: confirm..." note

CORRECTION

Backronym removed (title corrected to a neutral description of the R package). Printed "TODO" notes removed from this and the Fraga (2024) and Lopes (2024) entries. *Exact bibliographic metadata for Moreira (2022), Fraga (2024) and Lopes (2024) to be confirmed with supervisors.*

Erratum 6

PAGES / SECTIONS

Abstract; Ch.4 (Table 4.9 cross-algorithm, RVV feature-frequency Table, §"RVV top rules", M5Rules comparison); Ch.2; Appendix C

LOCATION

RVV LAI over-claim

SUBMITTED TEXT

LAI described as the **dominant RVV feature, "present in all 20 rules" / "20/20 rules"** (Abstract, Table 4.9, Appendix C, Ch.2, Ch.4).

CORRECTION

Not supported by the rule output (`rules_interpreted_rvv.csv`). Across the 20 RVV rules the **LAI family appears in 4/20** and **LAI at harvest specifically in 2/20**; the dominant RVV features are **heat-stress days (up to 50% of rules)** and **harvest soil water (25%)** — as the thesis's own frequency table (Table 4.9) already shows (LAI at harvest is rank 10, 10%). All "all 20 / 20/20 / dominant-LAI" statements corrected to 4/20 (family) or 2/20 (harvest); the Ch.2 example of a CarenR-specific feature changed from LAI (which all three algorithms select) to wet-days post-flowering. Extends the Appendix E correction (which had logged 4/20 but not propagated).

Note on nature of corrections. Items 1–5 are editorial/bibliographic and affect no result, table, test, or conclusion. Item 6 is a **factual correction to a feature-frequency description**: the RVV rule *counts* (20), MAEs, statistical tests, and conclusions are unchanged, but the claim that LAI is the dominant RVV feature was wrong — LAI appears in 4/20 rules, and heat-stress days and soil water are the dominant RVV features (consistent with the thesis's own Table 4.9 and Appendix E). The support-threshold value used in all computations was 20% throughout; item 4 corrects descriptive text that lagged the code, not the analysis itself.